

Dr. Alexandra Chen

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Austin, TX 78701, United States

OBJECTIVE

Seeking a faculty position in Machine Learning and Artificial Intelligence to leverage expertise in large-scale distributed learning systems, multimodal reasoning, and AI safety. Aiming to contribute to foundational research at the intersection of theoretical ML and real-world systems engineering in areas such as robustness, interpretability, and scalable inference.

EXPERIENCE

- **Google DeepMind** Jun 2023 — Sep 2023
Research Scientist Intern London, UK
 - Developed a distributed fine-tuning framework for 70B-parameter language models, reducing GPU memory overhead by 38% via gradient checkpointing and ZeRO-3 sharding
 - Implemented speculative decoding with adaptive draft models, achieving 2.4× inference throughput on TPUv4 clusters
 - Conducted ablation studies on chain-of-thought prompting strategies, identifying key factors driving 12-point accuracy gains on GSM8K
 - Presented findings at internal NeurIPS preparation workshop; work contributed to a published paper
- **Meta AI Research (FAIR)** May 2022 — Aug 2022
Research Engineer Intern Menlo Park, CA
 - Engineered a multimodal contrastive learning pipeline (image + text + audio) processing 50M samples/day on 256-GPU A100 cluster
 - Designed a data-efficient few-shot evaluation harness reducing benchmark annotation costs by 60%
 - Optimized CUDA kernels for attention computation, cutting training step time by 18% for 13B-parameter models
 - Co-authored internal technical report on self-supervised representation quality metrics
- **MIT CSAIL — Distributed ML Lab** Sep 2020 — May 2023
Graduate Research Assistant Cambridge, MA
 - Led research on Byzantine-fault-tolerant federated learning; published 2 papers at ICML and NeurIPS
 - Designed and maintained a 32-node GPU cluster testbed used by 8 research groups across 3 institutions
 - Mentored 4 undergraduate students, 2 of whom received NSF REU fellowships
 - Secured \$120K in compute credits from AWS and Google Cloud for lab research projects

EDUCATION

- **Massachusetts Institute of Technology** Sep 2019 — May 2024
Doctor of Philosophy in Electrical Engineering and Computer Science Cambridge, MA
 - GPA: 5.0/5.0
- **University of California, Berkeley** Aug 2015 — May 2019
Bachelor of Science in Computer Science and Mathematics Berkeley, CA
 - GPA: 3.97/4.00

SKILLS

- **Technical Skills:** Python, C++, CUDA, JAX, PyTorch, TensorFlow, Rust, Go, SQL, Bash, Distributed Systems, Federated Learning, LLM Fine-tuning, RLHF, Contrastive Learning, Bayesian Optimization, Linux, Docker, Kubernetes, AWS, GCP, Slurm, Git, LaTeX, Typst

PROJECTS

- **FaultlessFL: Byzantine-Robust Federated Learning Framework** Jan 2021 — Dec 2022
Principal Investigator, Lead Engineer [🔗](#)
 - Designed a novel aggregation protocol tolerating up to 40% Byzantine workers without accuracy degradation
 - Open-sourced framework adopted by 3 academic labs and 1 industry partner; 800+ GitHub stars
 - Supports PyTorch, JAX, and custom communication backends via a plugin architecture
- **MultiSense: Multimodal Reasoning Benchmark** Mar 2023 — Present
Co-Creator, Maintainer [🔗](#)
 - Curated 12,000 human-annotated cross-modal reasoning instances across vision, audio, and text
 - Evaluated 15 state-of-the-art models; revealed 23-point gap between human and best model performance

CERTIFICATIONS

- **AWS Certified Machine Learning — Specialty** Issued by: Amazon Web Services 2023
- **Google Cloud Professional ML Engineer** Issued by: Google Cloud 2022
- **NVIDIA Deep Learning Institute — Multi-GPU Training** Issued by: NVIDIA 2021

HONORS & AWARDS

- **NSF Graduate Research Fellowship** 2020
National Science Foundation
 - Awarded to 2,000 outstanding graduate researchers annually across all STEM fields; includes 3-year stipend and tuition coverage.
- **Best Paper Award — ICML 2022** 2022
International Conference on Machine Learning
 - Selected from 5,600 submissions for exceptional contribution to the federated learning track.
- **MIT Presidential Fellowship** 2019
Massachusetts Institute of Technology
 - Awarded to top 5% of incoming doctoral students.
- **Valedictorian** 2019
UC Berkeley EECS Department
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PATENTS & PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN

SUBMISSION, T=THESIS

- [J.1] A. Chen, R. Patel, L. Zhao (2023). **Byzantine-Robust Federated Learning via Adaptive Gradient Filtering**. *Journal of Machine Learning Research*.
- [C.3] A. Chen, S. Kim, J. Hoffmann (2023). **Speculative Decoding with Adaptive Draft Selection for Large Language Models**. *Advances in Neural Information Processing Systems (NeurIPS)*.
- [C.2] A. Chen, M. Torres, P. Singh, L. Zhao (2023). **MultiSense: A Cross-Modal Reasoning Benchmark for Foundation Models**. *NeurIPS Datasets and Benchmarks Track*.
- [C.1] A. Chen, R. Patel (2022). **Communication-Efficient Federated Learning under Byzantine Attacks**. *International Conference on Machine Learning (ICML)*.
- [S.1] A. Chen, K. Park, J. Hoffmann (2024). **Theoretical Guarantees for Multimodal Contrastive Representation Learning**. *Journal of Artificial Intelligence Research*.
- [P.1] A. Chen, R. Patel, L. Zhao (2023). **System and Method for Fault-Tolerant Gradient Aggregation in Distributed Neural Network Training**. *US Patent Office*.

LANGUAGES

English (Native), Mandarin Chinese (Fluent), French (Conversational)

INTERESTS

AI Safety and Alignment, Open-source software, Science communication, Distance running, Classical piano

VOLUNTEER EXPERIENCE

- **Instructor** Jun 2022 — Aug 2022
AI4ALL — Summer Program for Underrepresented Students 
 - Designed and taught a 4-week curriculum on ML fundamentals to 25 high school students
 - 85% of participants reported increased interest in pursuing STEM higher education
 - Developed open-source course materials used by 3 other AI4ALL chapters
- **Reviewer** 2021 — Present
NeurIPS, ICML, ICLR, AAAI
 - Reviewed 30+ papers annually across top-tier ML venues
 - Recognized as Outstanding Reviewer at NeurIPS 2022 and ICML 2023

REFERENCES

1. **Prof. Yael Tamir**
Associate Professor, EECS Department
Massachusetts Institute of Technology

Email: ytamir@csail.mit.edu

Phone: +1-617-555-0144

PhD Thesis Advisor

2. **Dr. James Hoffmann**

Research Director, Gemini Team

Google DeepMind

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Industry Research Mentor (Internship 2023)

3. **Prof. Rania Patel**

Professor, School of Engineering

Stanford University

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Phone: +1-650-555-0278

Collaborator — ICML 2022 Best Paper